

Alexander B. Kunin · Curriculum Vitae

Department of Mathematics
Creighton University
2500 California Plaza
Omaha, NE 68178

email: alexkunin@creighton.edu
office: (402) 280-2582
website: <https://sekunder.github.io/>
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Education and Employment

Employment	Creighton University Assistant Professor	Omaha, NE Jan 2023 –
	Baylor College of Medicine, University of Houston Postdoctoral Fellow <i>Mentors: Xaq Pitkow (BCM), Krešimir Josić (UH)</i>	Houston, TX Sep 2019 – Dec 2022
Education	Pennsylvania State University Ph.D. in Mathematics <i>Thesis: Properites and Applications of Convex Neural Codes</i> <i>Adviser: Vladimir Itskov</i>	State College, PA Aug 2014 – Aug 2019
	University of Nebraska – Lincoln M.S. in Mathematics	Lincoln, NE Aug 2012 – May 2014
	University of Alabama in Huntsville M.S. in Computer Science <i>Thesis: Self-Stabilizing Algorithms for Independence, Domination, and Coloring</i> <i>Adviser: Pete Slater</i>	Huntsville, AL Aug 2010 – Dec 2012
	Stony Brook University B.S. in Mathematics, minor in Computer Science	Stony Brook, NY Sep 2006 – May 2010
Additional	Allen Institute for Brain Science Summer Workshop on the Dynamic Brain	Friday Harbor, WA Aug 2016 – Sep 2016
	Center for Brains, Minds, and Machines Summer Course 2017	Woods Hole, MA Aug 2017 – Sep 2017
	Neuromatch Academy Online Course 2020	Jul 2020 – Jul 2020

Publications

Preprints	Adjoints of Morphisms of Neural Codes J. Geraci, A.B. Kunin, A. Seceleanu <i>Under Review</i> Preprint: https://arxiv.org/abs/2603.10837	2026
	Boolean Rank via Monomial Ideals J. Geraci, A.B. Kunin, A. Seceleanu <i>In revision</i> Preprint: https://arxiv.org/abs/2509.09570	2025

A network neuroscience for nanoscale connectomics 2025
 Q. Wang, G.J. Gutierrez, A.B. Kunin, J.T. Vogelstein, K.E. Bassler, K. Josić, R. Betzel
In revision.

Published The Oviposition Inhibitory Neuron is a potential hub of multi-circuit integration in the *Drosophila* brain 2025

R. Weber-Langstaff, P. Srivastava, A.B. Kunin, G.J. Gutierrez
eNeuro 12 (9)

Paper: <https://doi.org/10.1523/ENEURO.0123-25.2025>

Preprint: <https://doi.org/10.1101/2024.10.25.620362>

Functional connectomics reveals general wiring rule in mouse visual cortex 2025

Z. Ding, P.G. Fahey, S. Papadopoulos, E. Wang, B. Celii, C. Papadopoulos, A.B. Kunin, ..., X. Pitkow, J. Reimer, A.S. Tolias
Nature 640, 459–469

Paper: <https://doi.org/10.1038/s41586-025-08840-3>

Preprint: <https://doi.org/10.1101/2023.03.13.531369>

NEURD: automated proofreading and feature extraction for connectomics 2025

B. Celii, S. Papadopoulos, Z. Ding, P.G. Fahey, E. Wang, C. Papadopoulos, A.B. Kunin, ..., X. Pitkow, A.S. Tolias, J. Reimer
Nature 640, 487–496

Paper: <https://doi.org/10.1038/s41586-025-08660-5>

Preprint: <https://doi.org/10.1101/2023.03.14.532674>

Hierarchical Modular Structure of the *Drosophila* Connectome 2023

A.B. Kunin, J. Guo, K. Josić, X. Pitkow, K.E. Bassler
Journal of Neuroscience, Vol. 43, Issue 37

Paper: <https://doi.org/10.1523/JNEUROSCI.0134-23.2023>

Preprint: <https://doi.org/10.1101/2022.11.23.517722>

Code: <https://github.com/josiclab/flybrain-clustering>

Oriented Matroids and Combinatorial Neural Codes 2023

A.B. Kunin, C. Lienkaemper, Z. Rosen
Combinatorial Theory, Vol. 3, Issue 1

Paper: <https://doi.org/10.5070/C63160427>

Preprint: <https://arxiv.org/abs/2002.03542>

Hyperplane neural codes and the polar complex 2020

V. Itskov, A.B. Kunin, Z. Rosen

Appears in Topological Data Analysis: Abel Symposia, vol 15

Paper: https://doi.org/10.1007/978-3-030-43408-3_13

Preprint: <https://arxiv.org/abs/1801.02304>

Grant Proposals

Creighton University Haddix Research Fund : *Relating Microcircuit Structure and Dynamics in Neural Networks*
 Submitted to Dr. George F. Haddix President's Faculty Research Fund (Creighton). Not funded.

Feb 2024

NIH : *Relating multi-resolution connectome structure to the activity and function of neural networks*

Sep 2023

Submitted to BRAIN Initiative: Theories, Models and Methods for Analysis of Complex Data from the Brain (NIH). Not funded.

NIH T15LM007093: *NLM Training Program in Biomedical Informatics and Data Science*

Jan 2021 – Dec 2022

Funding for proposed project and curriculum of graduate courses and professional training in biomedical informatics and data science. Funded.

Service

Professional	Outside Committee Member, U. Nebraska – Lincoln Graduate Program	Fall 2025
	Organizer, AMS Special Session on Discrete, Algebraic, and Topological Methods in Mathematical Biology	Fall 2023
	UNL Preparing Future Faculty Mentor	Fall 2023
University	College of Arts and Sciences Faculty Senate	Spring 2026
	Neuroscience Curriculum Committee	Fall 2025
	Educating The Whole Person in an AI-Driven World: Session Leader, Technical Track	Summer 2025
	Pre-Health Advisory Committee	Spring 2025
	RSP Advisor	Fall 2024
Department	Goldwater Committee	Fall 2023
	Hiring Committee Fall 2024	Fall 2024
	Hiring Committee Fall 2023	Fall 2023
Reviewer	La Matematica	2026
	Natural Computing	2025
	PNAS	2024
	Cell Reports	2024
	Heliyon	2022
	Discrete and Computational Geometry	2022
	SIAM Journal on Applied Dynamical Systems	2020
	SIAM Journal on Applied Algebra and Geometry	2020

Presentations

Conference	AMS Spring Sectional Meeting – Special Session on Discrete and Algebraic Methods in Mathematical Biology. <i>Algebraic aspects of Boolean Matrix Factorization</i>	May 2025
	Nebraska-SE South Dakota MAA Section Meeting. <i>Neuroscience to Math and Back</i>	Apr 2025
	AMS Spring Sectional Meeting – Special Session on Applied Commutative Algebra. <i>Combinatorial Codes, Oriented Matroids, and Commutative Algebra</i>	Apr 2024

	SIAM Conference on Applied Algebraic Geometry – Minisymposium on Algebraic & Geometric Methods in Neural Coding. <i>Oriented Matroids and Combinatorial Neural Codes</i>	Aug 2021
	Neuromatch Conference 3.0. <i>Receptive field geometry shapes information content of the neural code</i>	Oct 2020
	SIAM Texas-Louisiana Section Meeting. <i>Low dimensional geometry of stimuli shapes the information content of a neural code</i>	Nov 2019
	SIAM Conference on Applied Algebraic Geometry – Minisymposium on Algebraic & Geometric Methods in Neural Coding. <i>Hyperplane Neural Codes and the Polar Complex</i>	Jul 2019
	AMS Spring Sectional Meeting – Special Session on Algebraic and Discrete Methods in Mathematical Biology. <i>Hyperplane Neural Codes and the Polar Complex</i>	Mar 2019
	AMS Fall Sectional Meeting – Special Session on Applied Algebraic Topology. <i>Hyperplane Neural Codes and the Polar Complex</i>	Sep 2018
	Joint Mathematics Meetings. <i>Hyperplane neural codes and the polar complex</i>	Jan 2018
	International Conference on Mathematical Neuroscience. <i>Low dimensional geometry of stimuli shapes the information content of a neural code</i>	May 2017
Poster	NITMB Converge Conference. <i>Beyond Convex Codes: Neural Rings in Boolean Matrix Factorization</i>	Aug 2025
	COSYNE Main Meeting. <i>Hierarchical Modular Structure of the Drosophila Connectome</i>	Mar 2023
	COSYNE Main Meeting. <i>Low dimensional geometry of stimuli shapes the information content of a neural code</i>	Feb 2017
Seminar	U. South Dakota Math Dept. Seminar. <i>From Brains to Binary Matrices</i>	Feb 2026
	UNL Biological Sciences Seminar. <i>Identifying network structure and connectivity rules in connectomes</i>	May 2025
	UNL Discrete Math Seminar. <i>The Convex Code Decision Problem (is probably not solvable)</i>	Nov 2023
	UNL Math Bio Seminar. <i>Mathematics and Neuroscience Sampler Platter</i>	Mar 2023
	Boston College Mathematics and Machine Learning Seminar. <i>Oriented Matroids and Combinatorial Neural Codes</i>	Oct 2021
Student	Creighton University Research Week. <i>Effects of Synaptic Degradation on Convergence and Memory Stability in a Hopfield Network (Winner, Best Undergraduate Natural/Physical Sciences Oral Presentation)</i> with Houda Tounkaoui*	Mar 2026
	Creighton University Research Week. <i>Circle-Forbidden Combinatorial Codes (Poster)</i> with Parker Abed*	Mar 2026

Joint Mathematics Meetings – PME Contributed Session on Research by Undergraduates. <i>Circle-Forbidden Combinatorial Codes</i> with Parker Abed*	Jan 2026
MAA meeting. <i>Circle-Forbidden Combinatorial Codes</i> with Parker Abed*	Nov 2025
AMS Fall Sectional Meeting – Saint Louis University Poster Session. <i>Circle-Forbidden Combinatorial Codes</i> with Parker Abed*	Oct 2025
Joint Mathematics Meetings – AMS-PME Undergraduate Student Poster Session. <i>How Complex can Convex Codes be?</i> with Parker Abed*	Jan 2025
Joint Mathematics Meetings – PME Contributed Session on Research by Undergraduates. <i>Improving Exponential Random Graph Models (ERGMs) Using Scaffolding for Enhanced Scalability and Reduced Degeneracy</i> with Nick Forbes*	Jan 2025
MAA Mathfest 2024 Undergraduate Poster Session. <i>Complexity of Convex Codes (working title)</i> with Parker Abed*	Aug 2024
MAA Mathfest 2024 Undergraduate Poster Session. <i>Scaffolded ERGMs (working title)</i> with Nick Forbes*	Aug 2024

* Undergraduate student presenting